

Topic 21

Biotechnology and genetic modification

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

21.1	Biotechnology and fermentation
21.2	Enzymes in industry
21.3	Genetic modification
21.4	Applications and evaluation

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Microorganisms in biotechnology

Microorganisms grow rapidly, use varied substrates and can produce enzymes, antibiotics and foods. Fermenters maintain temperature, pH, oxygen and nutrients while stirring distributes cells and heat. Sterile conditions prevent contamination and competition. Products are harvested and purified.

Examiner insight

Explain the purpose of each fermenter control rather than listing equipment.

2. Industrial enzymes

Pectinase increases fruit-juice yield and clarity; lactase makes lactose-free milk; biological washing powders contain proteases and lipases. Immobilised enzymes can be reused, are easy to separate from product and often tolerate wider conditions.

Examiner insight

Enzymes remain specific even in industrial processes.

3. Genetic modification

A desired gene is isolated, cut using a restriction enzyme and inserted into a vector such as a bacterial plasmid using ligase. The recombinant plasmid enters bacteria, which are selected, cloned and express the gene. Human insulin and some vaccine components are produced this way.

Examiner insight

Restriction enzymes cut DNA; ligase joins DNA fragments.

4. GM crops and evaluation

Genes can give pest resistance, herbicide tolerance, improved nutrition or stress tolerance. Benefits include yield and reduced insecticide use; concerns include gene flow, resistant pests, effects on non-target organisms, cost and farmer dependence. Evidence should be evaluated case by case.

Examiner insight

Avoid claiming all GM organisms have the same risks or benefits.

Worked application

Situation: A pest-resistant gm crop is widely planted.

Prediction: Insecticide use may fall but selection for resistant pests may increase.

Reason: The inserted gene protects plants, while strong selection favours rare resistant insects.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic 21 · Biotechnology and genetic modification

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which statement best defines biotechnology? [1]
 - A use of organisms, cells or enzymes to make useful products
 - B a controlled vessel for large-scale growth of microorganisms
 - C a small circular DNA molecule used as a vector
 - D an enzyme that cuts DNA at specific sequences
2. Which statement best defines fermenter? [1]
 - A a small circular DNA molecule used as a vector
 - B a controlled vessel for large-scale growth of microorganisms
 - C an enzyme that cuts DNA at specific sequences
 - D an enzyme that joins DNA fragments
3. Which statement best defines plasmid? [1]
 - A an enzyme that cuts DNA at specific sequences
 - B an enzyme that joins DNA fragments
 - C a small circular DNA molecule used as a vector
 - D use of organisms, cells or enzymes to make useful products
4. Which statement best defines restriction enzyme? [1]
 - A an enzyme that joins DNA fragments
 - B use of organisms, cells or enzymes to make useful products
 - C a controlled vessel for large-scale growth of microorganisms
 - D an enzyme that cuts DNA at specific sequences
5. Which statement best defines ligase? [1]
 - A an enzyme that joins DNA fragments
 - B use of organisms, cells or enzymes to make useful products
 - C a controlled vessel for large-scale growth of microorganisms
 - D a small circular DNA molecule used as a vector
6. Which statement about microorganisms in biotechnology is correct? [1]
 - A Study of fossils only.
 - B Microorganisms grow rapidly, use varied substrates and can produce enzymes, antibiotics and foods.
 - C A container that must always exclude oxygen.
 - D A bacterial protein coat.

7. Which statement about industrial enzymes is correct? [1]
- A A container that must always exclude oxygen.
 - B A bacterial protein coat.
 - C Pectinase increases fruit-juice yield and clarity; lactase makes lactose-free milk; biological washing powders contain proteases and lipases.
 - D An enzyme that joins dna fragments.
8. Which statement about genetic modification is correct? [1]
- A A bacterial protein coat.
 - B An enzyme that joins dna fragments.
 - C An enzyme that digests starch.
 - D A desired gene is isolated, cut using a restriction enzyme and inserted into a vector such as a bacterial plasmid using ligase.
9. Which statement about gm crops and evaluation is correct? [1]
- A Genes can give pest resistance, herbicide tolerance, improved nutrition or stress tolerance.
 - B An enzyme that joins dna fragments.
 - C An enzyme that digests starch.
 - D Study of fossils only.
10. Which prediction is most appropriate when a pest-resistant GM crop is widely planted? [1]
- A Study of fossils only.
 - B A bacterial protein coat.
 - C Insecticide use may fall but selection for resistant pests may increase.
 - D An enzyme that digests starch.

Section B · Structured questions

11 Biologists use precise vocabulary when studying biotechnology and genetic modification.

(a) Define biotechnology. [2]

(b) Define fermenter. [2]

(c) Distinguish plasmid from the incorrect idea: “a bacterial protein coat”. [2]

(d) State two important features of microorganisms in biotechnology. [2]

[Total: 8]

12 A pest-resistant gm crop is widely planted.

(a) State the most likely biological outcome. [2]

(b) Explain the outcome using a linked biological mechanism. [3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records relative fermenter yield under three conditions.

condition	relative fermenter yield
20 °C	3
30 °C	12
45 °C	2

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to investigate immobilised lactase at different flow rates.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of biotechnology and genetic modification to an unfamiliar context.

(a) Define restriction enzyme. [2]

(b) Explain the biological importance of genetic modification. [3]

(c) Correct this statement: “an enzyme that digests starch”.

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions • Multiple choice

- 1 A. Use of organisms, cells or enzymes to make useful products.
- 2 B. A controlled vessel for large-scale growth of microorganisms.
- 3 C. A small circular dna molecule used as a vector.
- 4 D. An enzyme that cuts dna at specific sequences.
- 5 A. An enzyme that joins dna fragments.
- 6 B. Explain the purpose of each fermenter control rather than listing equipment.
- 7 C. Enzymes remain specific even in industrial processes.
- 8 D. Restriction enzymes cut DNA; ligase joins DNA fragments.
- 9 A. Avoid claiming all GM organisms have the same risks or benefits.
- 10 C. The inserted gene protects plants, while strong selection favours rare resistant insects.

Solutions · Structured questions

11. (a) use of organisms, cells or enzymes to make useful products [2]. (b) a controlled vessel for large-scale growth of microorganisms [2]. (c) a small circular DNA molecule used as a vector; not a bacterial protein coat [2]. (d) Credit any two accurate features from the guide: Microorganisms grow rapidly, use varied substrates and can produce enzymes, antibiotics and foods. Fermenters maintain temperature, pH, oxygen and nutrients while stirring distributes cells and heat. Sterile conditions prevent contamination and competition. Products are harvested and purified. [2].

12. (a) Insecticide use may fall but selection for resistant pests may increase [2]. (b) The inserted gene protects plants, while strong selection favours rare resistant insects [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $12 - 2 = 10$ [2]. (b) The values change from 3 to 2, with the highest value 12. [2]. (c) Yield peaks near the microorganism enzyme optimum and falls when enzymes are denatured [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Flow rate [1]. (b) Glucose concentration in product [1]. (c) Same bead mass, lactose concentration, temperature and column [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Repeat and use a glucose calibration curve [1].

15. (a) an enzyme that cuts DNA at specific sequences [2]. (b) Award linked points from: A desired gene is isolated, cut using a restriction enzyme and inserted into a vector such as a bacterial plasmid using ligase. The recombinant plasmid enters bacteria, which are selected, cloned and express the gene. Human insulin and some vaccine components are produced this way. [3]. (c) Correct statement: ligase means an enzyme that joins DNA fragments [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Biology 0610 syllabus 2026-2028 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.